

RISK BY PROXY

How AI Wealth Management Tools Encode Demographic Bias and What Governance Frameworks Are Missing

A Policy Brief by Sabrina Rouabhi

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This brief draws on the author's professional background in wealth management operations and AI model evaluation to identify a governance gap at their intersection. It does not reflect the practices of any specific institution.

1. The Problem

When the Data Is the Bias

Robo-advisors were designed to remove human bias from financial advice. But when the data they learn from reflects decades of racial and economic inequality, removing the human does not remove the bias. It just makes it invisible.

Wealth management has been one of the last frontiers of financial services to adopt AI at scale. Today, algorithmic tools assess client risk tolerance, classify investors into service tiers, and generate personalized investment recommendations — all without human intervention. The promise is objectivity. The reality is more complicated.

These models do not learn from neutral data. They learn from historical financial records that encode decades of structural exclusion — redlining that concentrated wealth in certain zip codes, lending discrimination that denied generational wealth-building to entire communities, wage gaps that left women and minorities with smaller investable assets. The model is not programmed to discriminate. It is trained on discrimination. And it reproduces those patterns as if they were simply the natural order of financial risk.

The result is a system that appears objective because no human made the biased decision — but one that may systematically underserve minority clients, first-generation investors, and communities historically excluded from wealth management. Not by intention. By inheritance.

2. Why It Matters

The Stakes Are Not Hypothetical

This is not a theoretical concern. The pattern of algorithmic bias in financial services is documented, litigated, and growing.

In 2019, Apple Card's credit algorithm was found to offer significantly lower credit limits to women than men with identical or superior financial profiles. Goldman Sachs, which powered the algorithm, could not explain why. The New York Department of Financial Services launched an investigation. No definitive technical explanation was ever made public.

The Consumer Financial Protection Bureau has repeatedly warned that AI models in financial services can produce discriminatory outcomes without ever explicitly using protected characteristics. The mechanism is proxy variables — zip code, employment type, account size, investment history — inputs that are financially legitimate on their own but that collectively reconstruct race, gender, and national origin from the patterns encoded in historical data.

The wealth management sector is particularly exposed. The global robo-advisory market was valued at \$6.6 billion in 2023 and is projected to grow at 33% annually through 2030. As these tools scale, so does their potential for harm. A biased recommendation made to one client by a human advisor is an isolated incident. The same bias embedded in an algorithm is a systemic policy applied to millions simultaneously — invisibly, consistently, and at scale.

The communities most likely to be harmed are those already least served by traditional wealth management — first-generation investors, minority households, immigrants, and women. The promise of robo-advisors was democratization. Without governance, the reality may be discrimination at scale.

3. The Governance Gap

A Framework Built for Yesterday's Risk

Regulators have not stood still on AI. But the frameworks they have produced were designed for a different problem.

FINRA Notice 24-09, issued in 2024, addresses the use of AI in financial services with guidance on explainability, accuracy, and supervision of AI-generated communications. It asks whether a model's outputs can be understood and monitored. It does not ask whether those outputs are equitable across demographic groups.

The NIST AI Risk Management Framework offers a voluntary structure for identifying and mitigating AI risk across four functions — Govern, Map, Measure, and Manage. It acknowledges fairness and bias as risk categories. But it is voluntary, sector-agnostic, and provides no binding requirement for financial institutions to audit training data for encoded historical bias before deployment.

New York City's Local Law 144 requires bias audits for AI tools used in hiring decisions. It is the most concrete binding standard currently applied to algorithmic bias in the United States. It does not extend to financial services.

The gap is precise. No current regulatory framework requires a wealth management firm to ask two questions that sit at the foundation of equitable AI: what was this model trained on, and do its outputs treat demographically similar clients equitably? Algorithm governance and training data governance are treated as separate concerns — or not treated at all. Until both are required simultaneously, the bias that enters through historical data will continue to exit through investment recommendations.

4. A Proposed Framework

Two Pillars of Equitable AI in Wealth Management

Addressing bias in wealth management AI requires governance at two levels simultaneously — the data that trains the model and the outputs the model produces. The following framework proposes four audit criteria organized around these two pillars.

Pillar 1: Training Data Governance

Criterion 1 — Historical Bias Assessment

Before deployment, institutions must conduct a structured review of training data to identify variables that correlate with protected characteristics — race, gender, national origin, socioeconomic status. This includes documentation of data sources, time periods covered, and populations represented or excluded. Data that encodes historical discrimination must be flagged, adjusted, or removed before model training.

Criterion 2 — Representational Adequacy

Training datasets must be evaluated for demographic representation. Models trained predominantly on data from historically wealthy, white, male investors will learn those patterns as the norm. Institutions must document who is and is not represented in their training data and assess whether gaps create systematic disadvantage for underrepresented client groups.

Pillar 2: Output Equity Governance

Criterion 3 — Demographic Parity Testing

Institutions must periodically test whether clients with identical financial profiles but different demographic proxies receive equivalent risk tier assignments and investment recommendations. Where disparities are found, institutions must investigate, document, and correct them.

Criterion 4 — Explainability for Equity

Clients must be able to receive a plain-language explanation of how their risk profile was determined. Explainability requirements must go beyond technical accuracy — they must be assessable for fairness. An explanation that is technically correct but reflects biased weighting of proxy variables does not meet this standard.

These four criteria do not require institutions to eliminate the use of legitimate financial inputs. They require institutions to demonstrate that those inputs are being used equitably — and to produce evidence of that demonstration on a regular basis.

5. Call to Action

Governance Cannot Wait for the Next Scandal

The history of financial regulation is reactive. So is the history of technology regulation. Rules follow crises. Standards follow lawsuits. Frameworks follow headlines. In both domains, the pattern is the same: act after the harm is visible, not before.

Wealth management AI sits at the intersection of both. And right now, neither side is fully watching it.

Financial regulators understand the product but not always the model. Technology regulators understand the model but have no jurisdiction over the product. The result is a coordination gap where each assumes the other is handling it. Neither is.

This needs to change on two fronts simultaneously.

Regulators: Close the Coordination Gap

FINRA, the SEC, the CFPB, the FTC, and NIST should not be developing AI governance standards independently. A joint regulatory working group, modeled on existing coordination mechanisms like the Financial Stability Oversight Council, would allow financial and technology regulators to develop binding cross-domain standards together.

Specifically, this group should address:

- Mandatory training data documentation before model deployment
- Demographic parity testing as a standard component of model risk management
- Equity requirements that go beyond explainability and accuracy
- Clear jurisdictional accountability so institutions know who is responsible for enforcement

This is not a new idea in financial regulation. It is simply an idea that has not been applied to AI bias yet.

Financial Institutions: Don't Wait

Voluntary action is both possible and strategically rational. Institutions that conduct bias audits now are better positioned to shape the standards that will eventually be required of them. They are also better positioned to serve the next generation of wealth management clients, including first generation investors, minority households, and immigrant communities that traditional wealth management has historically underserved.

The client base is changing. The tools should reflect that.

Researchers: Shift the Focus

The literature on algorithmic bias in financial services has concentrated almost exclusively on credit and lending. Wealth management AI is critically understudied. Empirical research on training data composition, output equity, and demographic parity in robo-advisory systems is needed to ground this governance conversation in evidence.

The tools to detect this bias exist. The frameworks to require it do not. That gap is a choice. And choices can be reversed.

About the Author

Sabrina Rouabhi is an AI governance researcher and practitioner with a background in wealth management data analytics and AI model evaluation. She has worked across global financial institutions and AI-driven technology environments, specializing in system integrity, regulatory compliance, and responsible AI. She is currently developing EthicsEval, an applied bias evaluation framework for AI outputs in financial services.